



PRESIDENCY UNIVERSITY

Private University Estd. in Karnataka State by Act No. 41 of 2013
Italgapura, Rajankunte, Yelahanka, Bengaluru – 560064



**Design and Implementation of an AI-Powered Autonomous
Security Robotic Dog with Integrated Face Recognition**

A PROJECT REPORT

Submitted by

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Under the guidance of,

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**BACHELOR OF TECHNOLOGY
IN**

COMPUTER SCIENCE AND ENGINEERING

PRESIDENCY UNIVERSITY

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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING BONAFIDE CERTIFICATE

Certified that this report “Design and Implementation of an AI-Powered Autonomous Security Robotic Dog with Integrated Face Recognition” is a Bonafide work of Dashami S (20221CSE0106), Hemanth G (20221CSE0075), and Harsha Vardhana T (20221CSE0224), who have successfully carried out the project work and submitted the report for partial fulfilment of the requirements for the award of the degree of BACHELOR OF TECHNOLOGY in COMPUTER SCIENCE & ENGINEERING, during the academic year 2025–26.

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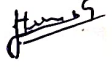
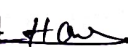
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PRESIDENCY SCHOOL OF COMPUTER SCIENCE AND ENGINEERING DECLARATION

We, the students of final year B.Tech in COMPUTER SCIENCE & ENGINEERING at Presidency University, Bengaluru, named Dashami S, Hemanth G, and Harsha Vardhana T, hereby declare that the project work titled **“Design and Implementation of an AI-Powered Autonomous Security Robotic Dog with Integrated Face Recognition”** has been independently carried out by us and submitted in partial fulfilment for the award of the degree of Bachelor of Technology in COMPUTER SCIENCE & ENGINEERING during the academic year **2025–26**. Further, the matter embodied in this project has not been submitted previously by anybody for the award of any Degree or Diploma to any other institution.

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Abstract

Autonomous home and institutional security has grown in importance as communities seek reliable monitoring solutions, especially when human supervision is limited or unavailable. The AI-Powered Autonomous Security Robotic Dog developed in this project is a low-cost, semi-autonomous surveillance companion designed to enhance safety by providing mobile, intelligent monitoring. Unlike fixed CCTV cameras that suffer from blind spots and lack responsiveness, the robotic dog integrates quadruped mobility, computer vision, facial recognition, motion detection, and behavioural intelligence to create an adaptive security assistant. The project was motivated by research showing that mobile robotic systems significantly improve environmental coverage, reduce security lapses, and enable real-time threat identification through AI-driven perception.

The report describes the robotic dog's mechanical and electronic architecture in detail. A 3D-printed quadruped frame powered by twelve servo motors enables smooth and stable locomotion across indoor environments, while a Raspberry Pi 5 coordinates servo actuation, sensor processing, and camera input. Real-time perception is achieved using a combination of motion detection, Haar Cascade face detection, and an advanced InsightFace model that generates facial embeddings stored in a FAISS vector database for rapid similarity searches. Environmental awareness is supported through calibrated servo movements and a synchronized control loop that allows the robot to scan, detect, and respond to dynamic changes. The onboard systems operate in conjunction with lightweight optimization modules to ensure reliable performance even under limited computational resources.

On the software side, the robotic dog integrates state-of-the-art computer vision and AI pipelines. YOLOv8 is used for object and face detection in real time, while the InsightFace framework provides robust identity recognition across varying lighting and angle conditions. A motion-based alert mechanism identifies suspicious activity and triggers recognition pipelines, allowing the robot to differentiate between familiar and unfamiliar individuals. These components work together to support an autonomous surveillance workflow capable of tracking movement, recognizing human faces, and issuing alerts during potential security breaches.